JATHIN KIRANI NAGARAJ

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Education

RV University Bengaluru,Karnataka
Bachelor of Technology (B.Tech) 2nd Year, **CGPA:** 8.45
Major: Artificial Intelligence & Machine Learning
Minor: Innovation, Entrepreneurship & Product Development
Elective: Photography
Expected Graduation: 2028

Deeksha K.M.W.A. Vidya Niketan school & Composite PU College Bengaluru,Karnataka
Board: State Board April2024

Experience

Trinarix Bengaluru,Karnataka
Technical Lead (Student Startup) June 2025 – Present

- Worked on the design and prototyping of MyOSA LifeSaver, a portable automated CPR system using ESP32, focusing on controlled chest compression, sensor-based feedback, and basic IoT monitoring.
- Currently building an AI-based sports training assistant for table tennis and boxing that analyzes practice data and provides improvement suggestions.
- Involved in problem identification, system design, prototyping, and iterative testing while continuing academic coursework.

RV University (ICSSR-Sponsored Project) Bengaluru, Karnataka
Research Assistant February 2025 – August 2025

- Contributed to a research project focused on improving mental and physical well-being of Bangalore City Police Officers through a holistic, tech-enabled program.
- Worked on frontend web development using React and Tailwind CSS.
- Served as UI/UX developer, tester, and debugger for the application.
- Curated digital content and handled video shooting and editing for the project platform.

Leadership & Activities

Student Clubs-RV University Bengaluru,Karnataka
Leadership Roles June 2025 – Present

- Held leadership roles across multiple student organizations.
- Served as President of a student club (currently inactive).
- Contributed to team coordination, planning, and execution of activities.

Projects

- **MyOSA LifeSaver – IoT-Based Automated CPR System**
  - Prototyped a portable automated CPR device using ESP32, real-time sensors, and IoT connectivity for monitoring and alerts.

- **PACT – AI & Digital Twin Based Proactive Network Congestion Prediction**
  - Built an AI-based proactive congestion prediction system using LSTM models trained on the IoT-23 dataset.
  - Integrated a Python-based Digital Twin with Isolation Forest for early anomaly and congestion detection.
  - Co-authored a research paper and applied for publication.

- **NeuroMatch AI – Intelligent Multi-Model Recruitment System**
  - Developed a multi-model AI recruitment framework using BERT-based NER, Random Forest, and Q-learning.
  - Achieved an average accuracy of 92.4%, outperforming traditional ATS systems.

- **IoT-Based Smart Emergency Vehicle Priority & Traffic Management System**
  - Implemented MQTT-based V2V and V2I communication to enable ambulance signal pre-emption and driver alerts.
  - Developed a hospital dashboard for real-time monitoring and ETA visualization.

- **Personal Financial Advisor App (In Development)**
  - Developing an AI-powered finance coaching app for Indian users featuring predictive spending analysis, NLP-based document parsing, and gamified financial literacy.

Skills & Interests

- **Technical:** C, HTML, CSS, Python (foundational scripting & data handling), Machine Learning fundamentals, Embedded systems prototyping (ESP32-based), IoT, Git, GitHub, VS Code, Arduino IDE, Cisco Packet Tracer, MySQL, Superbase
- **Product & Soft Skills:** Product thinking & design, analytical thinking, rapid prototyping mindset, problem-first approach, teamwork, leadership
- **Creative & Interests:** Photography (intermediate–advanced), video editing (DaVinci Resolve), game design, civic-tech innovation